07_bayesian_model

April 12, 2026

[1]:
```
import pandas as pd
import numpy as np
import pymc as pm
import arviz as az
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler

df = pd.read_csv('/content/drive/MyDrive/feature_engineered_real_estate.csv')
df.head()
```

[1]:
```
   TotalFinishedArea  LivingUnits  TotalAppraisedValue  SaleYear  SaleMonth  \
0             1092.0          1.0               187200      2024          7
1              936.0          1.0               192300      2022          4
2             1111.0          1.0               148400      2023         12
3             1684.0          1.0               197900      2021          1
4              900.0          1.0               170700      2021          9

   PriceRatio  AssrLandUse_APT FOUR  AssrLandUse_CONDOMINIMUM  \
0    1.383547                 False                     False
1    1.081643                 False                     False
2    0.842318                 False                     False
3    0.894391                 False                     False
4    1.101347                 False                     False

   AssrLandUse_MULTI DWLG  AssrLandUse_ONE FAMILY  AssrLandUse_THREE FAMILY  \
0                   False                    True                     False
1                   False                    True                     False
2                   False                    True                     False
3                   False                    True                     False
4                   False                    True                     False

   AssrLandUse_TWO FAMILY
0                   False
1                   False
2                   False
```

```
3                    False
4                    False
```

1 Step 1

[2]:
```python
# Check data
print(df.shape)
print(df.columns)
print(df.isnull().sum())
```

```
(3137, 12)
Index(['TotalFinishedArea', 'LivingUnits', 'TotalAppraisedValue', 'SaleYear',
       'SaleMonth', 'PriceRatio', 'AssrLandUse_APT FOUR',
       'AssrLandUse_CONDOMINIMUM', 'AssrLandUse_MULTI DWLG',
       'AssrLandUse_ONE FAMILY', 'AssrLandUse_THREE FAMILY',
       'AssrLandUse_TWO FAMILY'],
      dtype='object')
TotalFinishedArea           0
LivingUnits                 0
TotalAppraisedValue         0
SaleYear                    0
SaleMonth                   0
PriceRatio                  0
AssrLandUse_APT FOUR        0
AssrLandUse_CONDOMINIMUM    0
AssrLandUse_MULTI DWLG      0
AssrLandUse_ONE FAMILY      0
AssrLandUse_THREE FAMILY    0
AssrLandUse_TWO FAMILY      0
dtype: int64
```

[3]:
```python
# Define response and predictors
y_col = "PriceRatio"

x_cols = [
    "TotalFinishedArea",
    "LivingUnits",
    "TotalAppraisedValue",
    "SaleYear",
    "SaleMonth",
    "AssrLandUse_APT FOUR",
    "AssrLandUse_CONDOMINIMUM",
    "AssrLandUse_MULTI DWLG",
    "AssrLandUse_THREE FAMILY",
    "AssrLandUse_TWO FAMILY"
]
```

[4]:
```
# Split into training and test sets
df_train, df_test = train_test_split(df, test_size=0.2, random_state=6243)

print(df_train.shape)
print(df_test.shape)
```

```
(2509, 12)
(628, 12)
```

[5]:
```
# Standardize only the continuous predictors
cont_cols = [
    "TotalFinishedArea",
    "LivingUnits",
    "TotalAppraisedValue",
    "SaleYear",
    "SaleMonth"
]

scaler = StandardScaler()

df_train = df_train.copy()
df_test = df_test.copy()

df_train[cont_cols] = scaler.fit_transform(df_train[cont_cols])
df_test[cont_cols] = scaler.transform(df_test[cont_cols])

df_train.head()
```

[5]:
```
      TotalFinishedArea  LivingUnits  TotalAppraisedValue  SaleYear  \
625            0.987019     0.338207             0.647918 -0.012819
2623           1.197209     1.494284             0.447504 -0.882083
211           -1.160396    -0.817871            -1.449198 -0.012819
2552           0.374654     0.338207            -0.030803 -0.882083
1687          -1.122330    -0.817871            -1.241552 -0.012819

      SaleMonth  PriceRatio  AssrLandUse_APT FOUR  AssrLandUse_CONDOMINIMUM  \
625    0.670056    0.788397                 False                     False
2623   0.956213    1.022044                 False                     False
211    0.670056    1.122914                 False                     False
2552  -0.474571    0.669291                 False                     False
1687  -0.188414    0.988372                 False                      True

      AssrLandUse_MULTI DWLG  AssrLandUse_ONE FAMILY  \
625                    False                   False
2623                   False                   False
211                    False                    True
2552                   False                   False
```

```
1687                    False                    False

      AssrLandUse_THREE FAMILY  AssrLandUse_TWO FAMILY
625                      False                    True
2623                      True                   False
211                      False                   False
2552                     False                    True
1687                     False                   False
```

2 Step 2

[6]:
```
with pm.Model() as bayes_model:
    # Priors
    intercept = pm.StudentT("Intercept", nu=4, mu=0, sigma=10)

    w_TotalFinishedArea   = pm.Normal("w_TotalFinishedArea", mu=0, sigma=10)
    w_LivingUnits         = pm.Normal("w_LivingUnits", mu=0, sigma=10)
    w_TotalAppraisedValue = pm.Normal("w_TotalAppraisedValue", mu=0, sigma=10)
    w_SaleYear            = pm.Normal("w_SaleYear", mu=0, sigma=10)
    w_SaleMonth           = pm.Normal("w_SaleMonth", mu=0, sigma=10)

    w_APT_FOUR            = pm.Normal("w_APT_FOUR", mu=0, sigma=10)
    w_CONDOMINIMUM        = pm.Normal("w_CONDOMINIMUM", mu=0, sigma=10)
    w_MULTI_DWLG          = pm.Normal("w_MULTI_DWLG", mu=0, sigma=10)
    w_THREE_FAMILY        = pm.Normal("w_THREE_FAMILY", mu=0, sigma=10)
    w_TWO_FAMILY          = pm.Normal("w_TWO_FAMILY", mu=0, sigma=10)

    sigma = pm.HalfNormal("sigma", sigma=1)

    # Linear predictor
    mu = (
        intercept
        + w_TotalFinishedArea   * df_train["TotalFinishedArea"]
        + w_LivingUnits         * df_train["LivingUnits"]
        + w_TotalAppraisedValue * df_train["TotalAppraisedValue"]
        + w_SaleYear            * df_train["SaleYear"]
        + w_SaleMonth           * df_train["SaleMonth"]
        + w_APT_FOUR            * df_train["AssrLandUse_APT FOUR"]
        + w_CONDOMINIMUM        * df_train["AssrLandUse_CONDOMINIMUM"]
        + w_MULTI_DWLG          * df_train["AssrLandUse_MULTI DWLG"]
        + w_THREE_FAMILY        * df_train["AssrLandUse_THREE FAMILY"]
        + w_TWO_FAMILY          * df_train["AssrLandUse_TWO FAMILY"]
    )

    # Likelihood
```

```
    y_obs = pm.Normal("PriceRatio", mu=mu, sigma=sigma,␣
↪observed=df_train[y_col])

    # Posterior sampling
    trace_bayes = pm.sample(
        draws=1000,
        tune=1000,
        chains=2,
        random_seed=6243,
        target_accept=0.9,
        return_inferencedata=True
    )
```

Output()

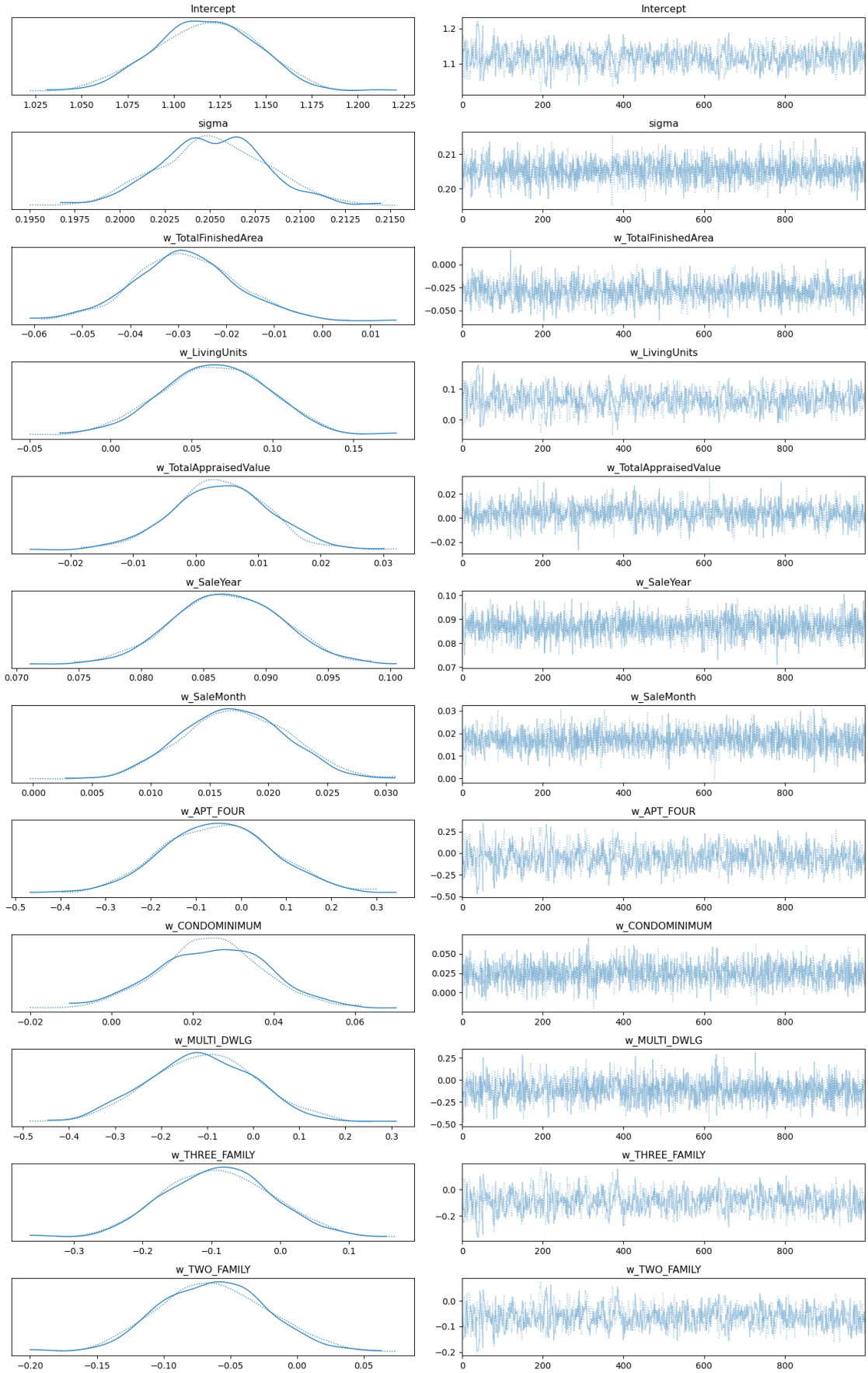
3 Step 3

[7]:
```
# Posterior summaries

# Summary table
summary = az.summary(trace_bayes, round_to=4)
print(summary)
```

```
                        mean      sd  hdi_3%  hdi_97%  mcse_mean  mcse_sd  \
Intercept             1.1177  0.0279  1.0681   1.1702     0.0011   0.0010
sigma                 0.2052  0.0030  0.1997   0.2109     0.0001   0.0001
w_APT_FOUR           -0.0493  0.1225 -0.2807   0.1780     0.0045   0.0038
w_CONDOMINIMUM        0.0248  0.0130 -0.0014   0.0478     0.0003   0.0003
w_LivingUnits         0.0646  0.0327  0.0060   0.1274     0.0013   0.0011
w_MULTI_DWLG         -0.1101  0.1194 -0.3334   0.1084     0.0031   0.0026
w_SaleMonth           0.0172  0.0044  0.0089   0.0248     0.0001   0.0001
w_SaleYear            0.0869  0.0043  0.0790   0.0949     0.0001   0.0001
w_THREE_FAMILY       -0.0903  0.0780 -0.2353   0.0531     0.0030   0.0028
w_TWO_FAMILY         -0.0625  0.0404 -0.1337   0.0146     0.0016   0.0014
w_TotalAppraisedValue  0.0040  0.0077 -0.0107   0.0183     0.0002   0.0002
w_TotalFinishedArea   -0.0286  0.0106 -0.0488  -0.0082     0.0003   0.0002

                       ess_bulk   ess_tail   r_hat
Intercept              667.6101   628.2454  1.0041
sigma                 1891.8965  1462.9899  0.9995
w_APT_FOUR             735.1860   764.2407  1.0041
w_CONDOMINIMUM        1825.5562  1502.5887  1.0040
w_LivingUnits          667.9631   696.9564  1.0040
w_MULTI_DWLG          1527.3372  1441.2888  1.0015
w_SaleMonth           2020.9575  1475.3449  1.0013
```

```
w_SaleYear              2013.5927  1480.6548  1.0003
w_THREE_FAMILY           673.4236   679.8868  1.0044
w_TWO_FAMILY             673.7189   605.1792  1.0049
w_TotalAppraisedValue   1185.3699  1328.4052  1.0027
w_TotalFinishedArea     1380.6601  1292.8358  1.0010
```

[8]:
```
az.plot_trace(
    trace_bayes,
    var_names=[
        "Intercept", "sigma",
        "w_TotalFinishedArea", "w_LivingUnits", "w_TotalAppraisedValue",
        "w_SaleYear", "w_SaleMonth",
        "w_APT_FOUR", "w_CONDOMINIMUM", "w_MULTI_DWLG",
        "w_THREE_FAMILY", "w_TWO_FAMILY"
    ],
    figsize=(14, 22)
)
plt.tight_layout()
plt.show()
```


Intercept
sigma
w_TotalFinishedArea
w_LivingUnits
w_TotalAppraisedValue
w_SaleYear
w_SaleMonth
w_APT_FOUR
w_CONDOMINIMUM
w_MULTI_DWLG
w_THREE_FAMILY
w_TWO_FAMILY